session 12 matrix multiplication and regression math intuition

Matrix Dot Products, Linear Equation Predictive Models & Recommendation Intuition

1. Instagram Engagement 2x2 Matrix Multiplication

Scenario: Multiply two 2x2 matrices representing Instagram post metrics (likes & comments across 2 days) using Python and NumPy.

```
import numpy as np # Matrix A: Post 1 and Post 2 engagement metrics [Likes, Comments] on Day 1 & Day 2 matrix_A = np.array([ [120, 15], # Day 1: 120 Likes, 15 Comments [200, 25] # Day 2: 200 Likes, 25 Comments ]) # Matrix B: Weight/Transformation matrix for engagement scoring matrix_B = np.array([ [2, 1], [5, 3] ]) # Matrix multiplication (Dot Product) result_matrix = np.matmul(matrix_A, matrix_B) # Or matrix_A @ matrix_B print("Matrix A:\n", matrix_A) print("Matrix B:\n", matrix_B) print("Resulting Product Matrix:\n", result_matrix)
```

MATRIX DOT PRODUCT OUTPUT

Resulting Matrix:
[[315, 165]
[525, 275]]

2. Linear Regression Prediction Function

Task: Create a Python function `linear_regression_predict(x, m, c)` that evaluates the linear regression line `y = mx + c`.

```
def linear_regression_predict(x, m, c): """Calculates predicted output y given input x, slope m, and intercept c.""" y = (m * x) + c return y # Function test case test_x, test_m, test_c = 5, 2, 3 predicted_y = linear_regression_predict(test_x, test_m, test_c) print(f"For x={test_x}, m={test_m}, c={test_c} -> Predicted y = {predicted_y}")
```

FUNCTION VERIFICATION

Predicted y: **13**

3. Zomato Delivery Time Predictions

Model: Equation y = 4x + 10 (where x = distance in km, y = time in minutes). Predict delivery times for 2km, 5km, and 8km.

```
distances = [2, 5, 8] # in kilometers m_slope = 4 # 4 minutes per kilometer
c_intercept = 10 # 10 minutes base preparation time print("--- Zomato Delivery Time
Predictions ---") for dist in distances: pred_time = linear_regression_predict(dist,
m_slope, c_intercept) print(f"Distance: {dist} km -> Predicted Delivery Time: {pred_time}
mins")
```

PREDICTION RESULTS

- **2 km distance:** (4 × 2) + 10 = **18 minutes**
- **5 km distance:** (4 × 5) + 10 = **30 minutes**
- **8 km distance:** (4 × 8) + 10 = **42 minutes**

4. Intuitive Interpretation: Flipkart Product Price Prediction

Context: Predicting product price (y) based on customer rating (x).

Slope (m) Interpretation

The **slope (m)** represents the rate of change in product price for every 1-unit increase in rating. It indicates how much extra premium customers pay per rating point; for instance, a slope of +500 means each additional star adds ₹500 to expected price.

Intercept (c) Interpretation

The **intercept (c)** represents the baseline price of a product when its rating score is zero. It reflects the fundamental floor cost of materials, packaging, and platform listing fees before factoring in customer satisfaction ratings.

5. Matrix Multiplication in Spotify Recommendation Systems

Matrix Factorization & Latent Feature Embeddings

Spotify represents millions of users and songs as a massive **User-Item interaction matrix**. Machine learning models use matrix factorization (like SVD or ALS) to decompose this table into two lower-dimensional matrices: a *User-Feature matrix* and a *Song-Feature matrix*. Matrix multiplication recombines these vectors via dot products to predict unobserved listening preference scores, enabling real-time personalized playlist recommendations.